

LACHLAN C. HAYDON

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EDUCATION

Loyola University Chicago

B.S. Physics | B.B.A. Information Systems & Business Analytics | GPA: 3.57

Chicago, IL
May 2026

PUBLICATIONS AND PRESENTATIONS

L. Haydon, R. Abbasi, and the Telescope Array collaboration, "Studying the Optical Emission of Lightning at the Telescope Array Detector," 2026 American Physical Society Global Summit, March 2026.

R. Abbasi, et al., "Observation of a Downward Terrestrial Gamma-Ray Flash Associated with Dart Leaders of a Negative Cloud-to-Ground Lightning Flash," Submitted to *Journal of Geophysical Research - Atmospheres*.

RELEVANT EXPERIENCE

qNoise – Quantum Noise Characterization & Benchmark (*Independent Project*)

2026

lachlan.site/qNoise

- Built an semi-automated pipeline to characterize qubit noise on IBM Quantum's `ibm_fez` (156-qubit Heron processor), measuring T1 energy relaxation, T2 Hahn echo coherence, and readout assignment error across 45 hardware observations over 5 days.
- Benchmarked Infleqtion's Superstaq compiler against Qiskit (optimization level 3) on identical circuits run on real hardware; found Superstaq achieved consistently higher Hellinger fidelity on QFT-4q (avg 0.9934 vs. 0.9886) despite producing a deeper circuit, by reducing two-qubit gate count.

Research Assistant – Computational Physics & Data Integration

January 2025 – Present

Loyola University Chicago | Advisor: Dr. Rasha Abbasi

Chicago, IL

- Integrated and synchronized data streams from 5+ geographically distributed detector instruments, including optical, electromagnetic, and particle detection systems, with microsecond-precision timing alignment across 300+ GB of experimental datasets.
- Designed GUI-based analysis platform by directly assessing researcher needs and translating experimental requirements into practical software solutions, enabling team members without computational backgrounds to perform complex analysis independently; platform now serves as standard tool for all incoming team members.
- Developed visualization dashboards and publication-quality plots communicating complex temporal patterns to technical and non-technical stakeholders; findings presented at American Physical Society Global Summit (March 2026).
- Awarded Mulcahy Fellowship for research excellence; coordinate analysis methods across research team and train team members on computational techniques and platform usage.
- Co-authored research paper submitted to peer-reviewed *Journal of Geophysical Research - Atmospheres*.

Summer Intern – Risk Management Analyst

June 2025 – August 2025

Marsh

Chicago, IL

- Served as technical point of contact for 85+ commercial insurance placements.
- Transformed policy data, exposure metrics, and coverage analysis into compelling narratives through technical deliverables including policy summaries, exposure schedules, and coverage comparisons for executive decision-making.

Lead Writing Center Tutor

September 2023 – December 2024

Loyola University Chicago

Chicago, IL

- Promoted to leadership role among 60+ tutors; conducted performance evaluations, developed training materials, and mentored incoming team members on effective communication strategies.
- Demonstrated ability to translate complex concepts into accessible language for diverse audiences.

Physics Tutor

January 2024 – Present

Loyola University Chicago STEM Center

Chicago, IL

LEADERSHIP & AWARDS

Damen Award of Excellence (2026) | Mulcahy Fellowship (2025) | Dean's List Sigma Pi Sigma (Physics Honor Society)

Vice-President, Loyola Physics Club | Safety Officer (Exec. Board), Loyola Club Swim Team | Lead Tutor, Loyola Writing Center

SKILLS

Python: NumPy, SciPy, Pandas, Plotly, Tkinter, Qiskit, qiskit-ibm-runtime, SQLAlchemy

Other: SQL, R, C++ (ROOT), Git, Tableau, Statistical Modeling, Technical Writing, Microsoft Office, Claude Code

ADDITIONAL EXPERIENCE

Lifeguard & Swim Coach (*Various Locations, Chicago & Bay Area*)

May 2019 – Present

Assistant Grounds Keeper, Saint Simon Parish & School, Los Altos, CA

Summers 2023, 2024